

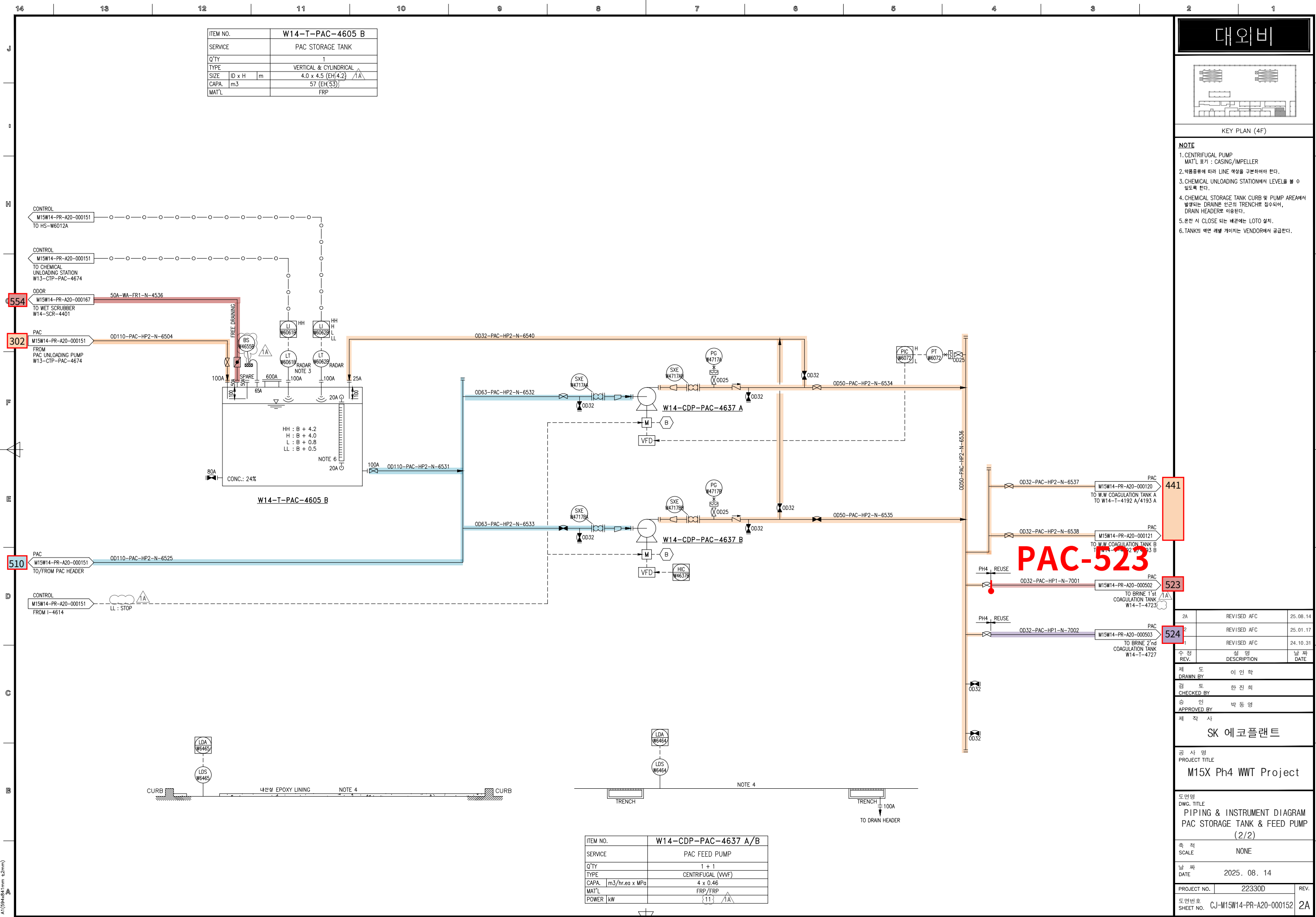
DRAWING LIST FOR TEST PACKAGE

PAC-523

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	523	R2FA	OD32-PAC-HP2-E-7001-01	1 OF 1	1	CJ-M15W14-PR-A20-000152		
2	523	R2FA	OD32-PAC-HP2-E-7004-01	1 OF 1	1	CJ-M15W14-PR-A20-000502		
3	523	2FA	OD32-PAC-HP2-E-7001-01	1 OF 1	1	CJ-M15W14-PR-A20-000152		
4								
5								
6								
7								
8								
9								
10								
11								
12								



ITEM NO.	W14-T-PAC-4605 B		
SERVICE	PAC STORAGE TANK		
Q'TY	1		
TYPE	VERTICAL & CYLINDRICAL		
SIZE	ID x H	m	4.0 x 4.5 (EH/4.2) / 1A
CAPA.	m3		57 (EH/53)
MAT'L	FRP		

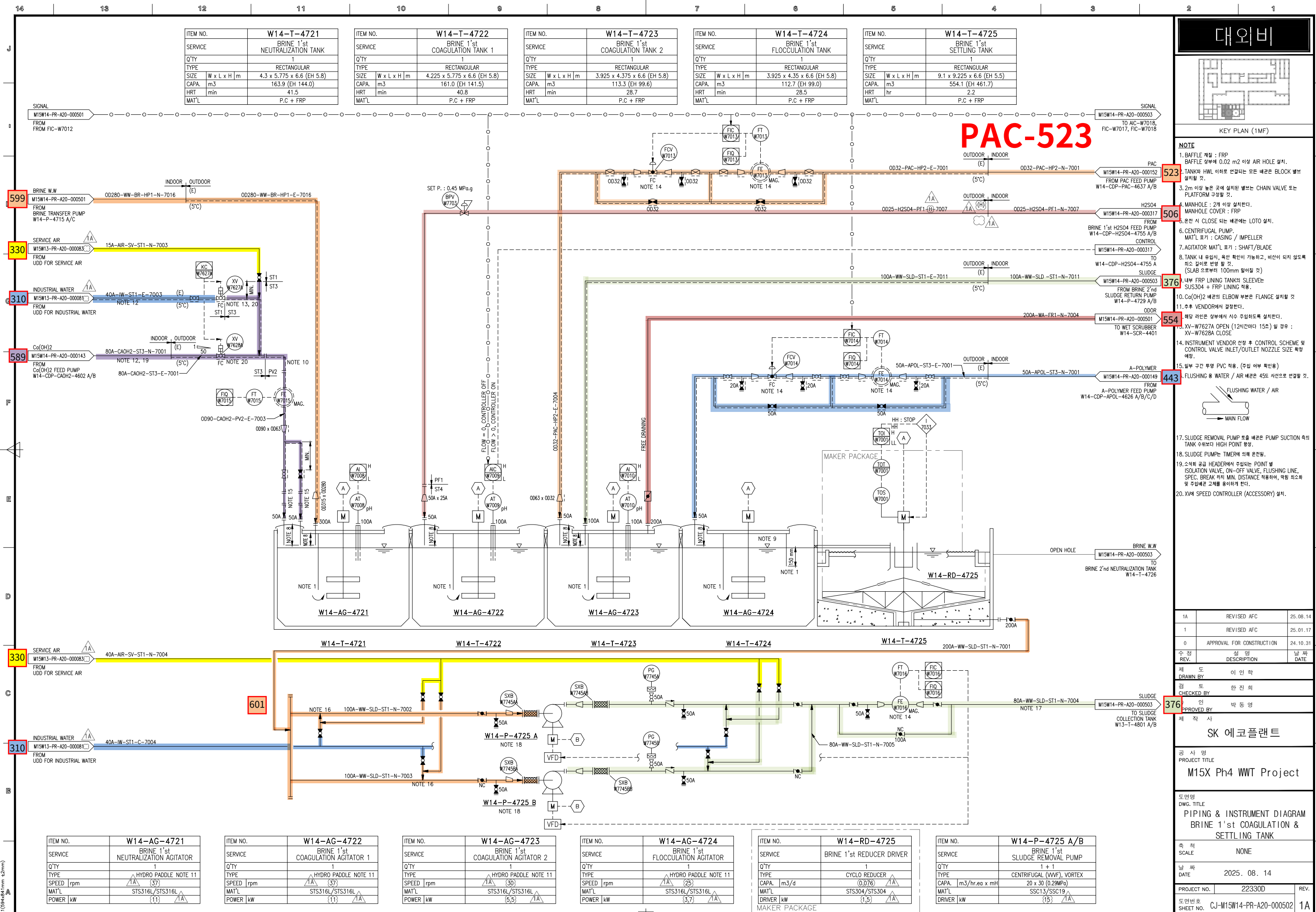
ITEM NO.		W14-CDP-PAC-4637 A/B	
SERVICE		PAC FEED PUMP	
Q'TY		1 + 1	
TYPE		CENTRIFUGAL (VVF)	
CAPA.	m3/hr.ea x MPa	4 x 0.46	
MAT'L		FRP/FRP	
POWER	kW	{ 11 }	1A

대외비

KEY PLAN (4F)

NOTE
1. CENTRIFUGAL PUMP
MAT'L 표기 : CASING/IMPELLER
2.약품종류에 따라 LINE 색상을 구분하여야 한다.
3. CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
4. CHEMICAL STORAGE TANK CURB 및 PUMP AREA에서 발생되는 DRAIN은 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이송된다.
5. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
6. TANK의 역전 레벨 경계치는 VENDOR에서 공급한다.

수정 REV.	실명 DESCRIPTION	날짜 DATE
2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
	REVISED AFC	24.10.31
수정 REV.	실명 DESCRIPTION	날짜 DATE
제 도 이 인 학		
검 토 한 진 회		
승 인 박 동 명		
제 작 사		
SK 에코플랜트		
공 사 명 PROJECT TITLE	M15X Ph4 WWT Project	
도면명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM PAC STORAGE TANK & FEED PUMP (2/2)	
축 척 SCALE	NONE	
날 짜 DATE	2025. 08. 14	
PROJECT NO.	22330D	REV. A
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000152	2A



대외비

KEY PLAN (1MF)

NOTE

1. Baffle 제형 : FRP
Baffle 상부에 0.02 m2 이상 AIR HOLE 설치.

2. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.

3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.

4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP

5. 온전 시 CLOSE 되는 배관에는 LOTO 설치.

6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER

7. AGITATOR MAT'L 표기 : SHAFT/BLADE

8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 반영 할 것.
(SLAB으로부터 100mm 떨어질 것)

9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.

10. Co(OH)2 배관의 ELBOW 부분은 FLANGE 설치할 것

11. 주후 VENDOR에서 결정한다.

12. 해당 라인은 상부에서 시수 주입하도록 설치한다.

13. XV-W7627A OPEN (12시간마다 15초) 일 경우 : XV-W7628A CLOSE

14. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정 예정.

15. 일부 구간 투명 PVC 적용. (주입 여부 확인용)

16. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.

17. SLUDGE REMOVAL PUMP 배관 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 형성.

18. SLUDGE PUMP는 TIMER에 의해 운전됨.

19. 소액의 공금 HEADER에서 주입되는 POINT 별 ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE, SPEC. BREAK 까지 MIN. DISTANCE 적용하여, 역침 침소화 및 주입배관 고제를 용이하게 한다.

20. XVM SPEED CONTROLLER (ACCESSORY) 설치.

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.10.31
수정	실명	날짜
REV.	DESCRIPTION	DATE

제 도

이 인 학

검 토

한 진 회

CHECKED BY

인

박 동 명

APPROVED BY

제 작 사

SK 에코플랜트

공 사 명

PROJECT TITLE

M15X Ph4 WWT Project

도면명

DWG. TITLE

PIPING & INSTRUMENT DIAGRAM

BRINE 1'st COAGULATION & SETTLING TANK

속 척

SCALE

NONE

날 짜

DATE

2025. 08. 14

PROJECT NO.

22330D

REV.

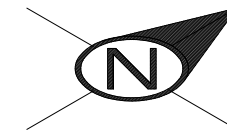
A

도면번호

SHEET NO.

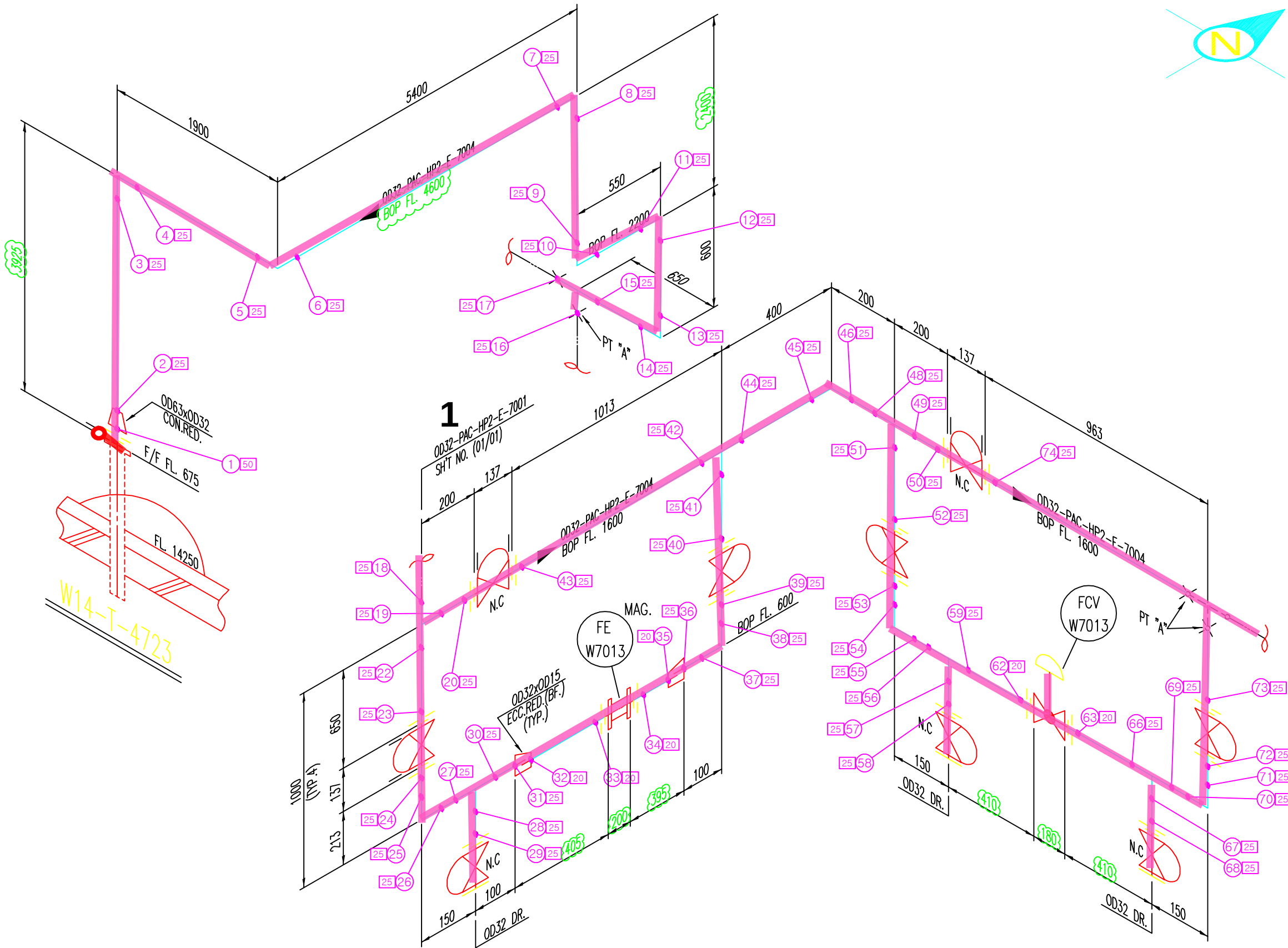
CJ-M15W14-PR-A20-000502

1A



1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

	CONTRACTOR 	PROJECT NAME M15X Ph4 Project	
和	COMMODITY PAC	JOB NO. 22330D	
和	LINE NO. 0D32-PAC-HP2-E-7001	SH'T NO. 01 / 01	REV. 1
D			



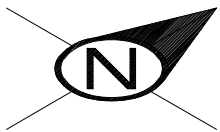
<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는
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2. (*) FIELD ADJUST

SK ecoplant			
ITEM	SIZE	Q'TY	DESCRIPTION
HP2			
P	OD25	1.8M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD32	22.8M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
E90	OD32	11EA	ELBOW 90 DEG, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD32	7EA	TEE EQUAL, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
RC	OD63xOD32	1EA	REDUCER CON, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
RE	OD32xOD15	2EA	REDUCER ECC, HDPE PE100, PE, BUTT FUSION, PN16/SDR11
VD	OD32	8EA	DIAPHRAGM VALVE, KS 10K, FF FLC'D, BODY-PP / TRIM-EPDM+PTFE DIAPHRAGM, BB, HANDWHEEL
F	OD25	4EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-ST304 / STUB END-HDPE PE100
F	OD32	15EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-ST304 / STUB END-HDPE PE100
F	OD63	1EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-ST304 / STUB END-HDPE PE100
FB	OD32	3EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-ST304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD25	4EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD32	15EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD63	1EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD32	3EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B	OD25	4SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	OD32	18SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	OD63	15SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
LOTO	OD32	5EA	LOTO
INC	25tXOD25	1.8M	관 보존재(고무벌포) + 알루미늄 CASING
INC	25tXOD32	22.8M	관 보존재(고무벌포) + 알루미늄 CASING
INC	25tXOD63	0.1M	관 보존재(고무벌포) + 알루미늄 CASING

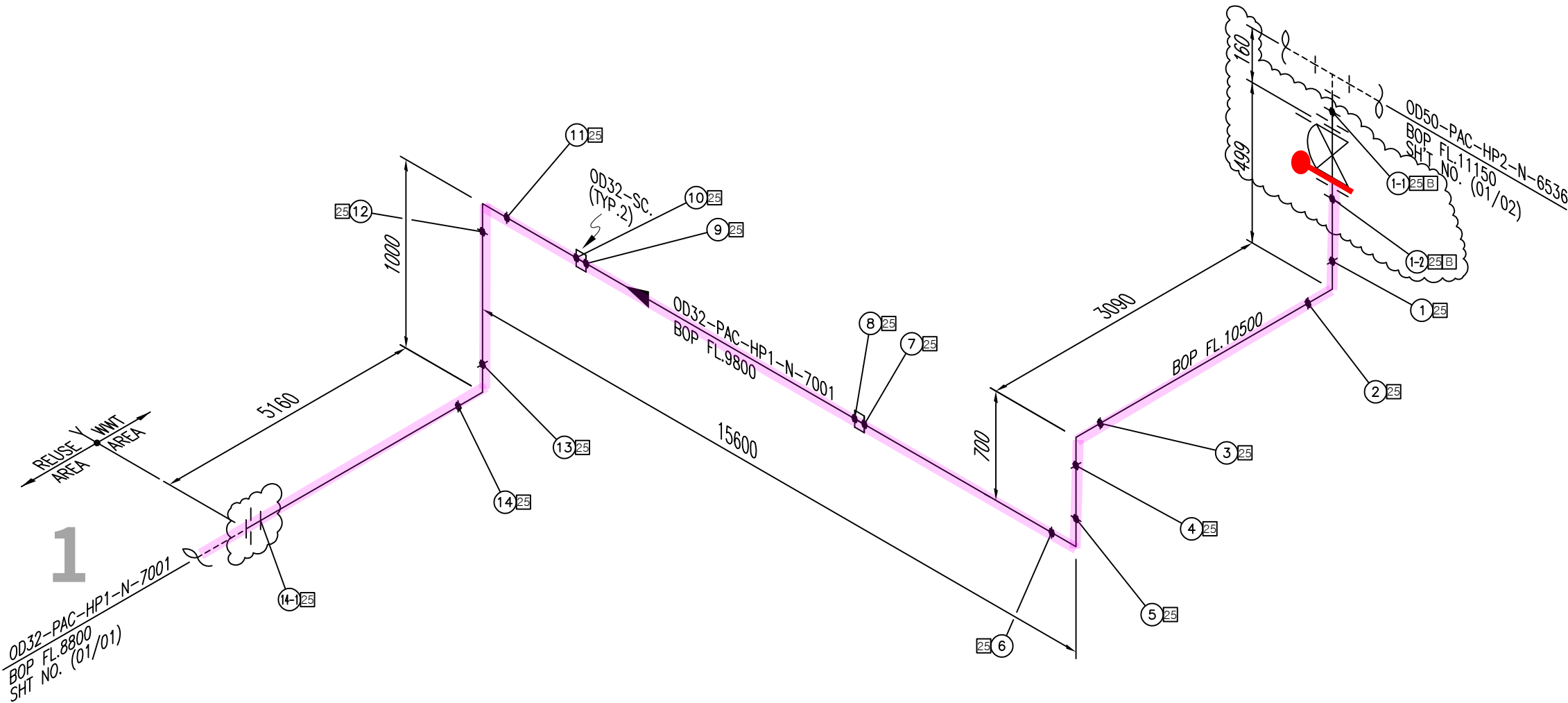
* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14								
	THICK.	25	DESIGN	6.4	60	UNIT	R2FA								
			OPERATING	4	AMB	REF. PLAN DWG.			25.03.31	REVISED AS MARKED	황래중	안남후	박동영		
						CJ-M15W14-PI-A30-020015-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영		
						REF. P. & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D		
						CJ-M15W14-PR-A20-000502									

CLIENT		2	
SK hynix			
CONTRACTOR			
SK ecoplant			
COMMODITY			
PAC			
LINE NO.			
OD32-PAC-HP2-E-7004			



ITEM	SIZE	Q'TY	DESCRIPTION
HP1			
P	OD32	26.2M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
E90	OD32	5EA	ELBOW 90 DEG, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
SC	OD32	2EA	SOCKET, HDPE PE100, SOCKET FUSION, PN16/SDR11
F	OD32	3EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STSS304 / STUB END-HDPE PE100
G	OD32	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD32	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
VD	OD32	2EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-PP TRIM-EPDM+PTFE DIAPHRAGM, BB, HANDWHEEL



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VBLO/VB1/VB2/VB2LO * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT		3	
CONTRACTOR		PROJECT NAME	
SK ecoplant		M15X Ph4 Project	
COMMODITY		JOB NO.	
PAC		22330D	
LINE NO.		SH'T NO.	REV.
OD32-PAC-HP1-N-7001		01 / 01	0

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14
	THICK.	-	DESIGN	1	60	UNIT	2FA
			OPERATING	ATM	AMB	REF. PLAN DWG.	
						CJ-M15W14-PI-A30-020011-H	
						REF. P & ID	
						CJ-M15W14-PR-A20-000152	

△	24.12.20	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D